Chapter 26: INT 16H, Keyboard handler

Use

Keyboard input and status.

Entry

AH selects the function required, as follows.

Exit

The AX register returns results as shown in the following descriptions. All other registers are preserved. The flags are corrupted.

Comments

The IBM PC BIOS provides the following functions:

00H	Get key	05H	Write to keyboard buffer
01H	Get status	10H	Extended read keystroke
02H	Get shift status	11H	Get extended status
03H	Set key repeat	12H	Get extended shift status

The **Hunter 16** BIOS provides identical functions and also supports a number of extensions, providing access to the following facilities:

20H	Set temporary shift
21H	Control shift keys
22H	Control [Ctrl][Alt][DEL]
23H	Control emergency breakout
24H	Redefine key codes
25H	Reset keyboard
26H	Control keyclick
27H	Control screen dump area
28H	Reserved
29H	Get key repeat
2AH	Control key repeat
2BH	Redefine key scan codes
2CH	Redefine PAW key codes
2DH	Control [Ctrl-C] and [Ctrl-Break]

(Functions 3 and 5 are compatible with the IBM PS/2 BIOS.)

Get key

Entry AH=00H

Exit AH=scan code
AL=character

Description Reads and then removes a keystroke from the keyboard input buffer. (If there is none, the function waits for one.)

The ASCII code is returned in AL. The key scan code is returned in AH. Key codes and scan codes are described in Appendix E.

Get status

Entry AH=01H

Exit ZF=flag
AX=character

Description Checks if there are any characters in the keyboard buffer.

If there is a character, ZF=0 and its ASCII code is returned in AX.

If there is no character, ZF=1.

NOTE: This function does not remove the character from the buffer.

Get shift status

Entry AH=02H

Exit AL=status

Description Checks the status of the modifier keys and returns the result as separate bits in AL, as follows:

AL	Significance
D7=1	INS mode
D6=1	CAPS LOCK mode
D5=1	NUM LOCK mode
D4=1	SCROLL LOCK mode
D3=1	[Alt] pressed
D2=1	[Ctrl] pressed
D1=1	Left [Shift] pressed
D0=1	Right [Shift] pressed

Set key repeat

Entry

AH=03H
AL=05H
BL=typematic rate (0..1FH)
BH=delay (0..3)

Exit

Returns nothing

Description

Sets the delay before keys repeat, and the rate at which they repeat. The rate (in characters per seconds) is set by BL:

BL	Rate	BL	Rate	BL	Rate	BL	Rate
0	30.0	8	15.0	10	7.5	18	3.7
1	26.7	9	13.3	11	6.7	19	3.3
2	24.0	A	12.0	12	6.0	1A	3.0
3	21.8	B	10.9	13	5.5	1B	2.7
4	20.0	C	10.0	14	5.0	1C	2.5
5	18.5	D	9.2	15	4.6	1D	2.3
6	17.1	E	8.6	16	4.3	1E	2.1
7	16.0	F	8.0	17	4.0	1F	2.0

The delay before a key repeats is set by BH:

BH	Delay
0	250mSec
1	500mSec
2	750mSec
3	1000mSec
4..FFH	Reserved

Write to buffer

Entry
AH=05H
CH=scan code
CL=ASCII code

Exit
AL=0 (success) or AL=1 (failed, buffer full).

Description
Inserts a 16-bit key token in the keyboard input buffer, exactly as if the user had pressed the corresponding key. Scan codes and ASCII codes are summarised in Appendix E.

Extended get key

Entry
AH=10H

Exit
AL=ASCII code
AH=scan code

Description
This function is exactly the same as function 00H. Scan codes and ASCIİ codes are summarised in Appendix E.

Extended get status

Entry
AH=11H

Exit
ZF=flag
AL=ASCII code
AH=scan code

Description
Checks if there are any characters in the keyboard buffer. If there is a character, ZF=0 and its ASCII code and scan code are returned in AL and AH. If there is no character, ZF=1.

NOTE: This function does not remove the character from the buffer.

Extended get shift status

Entry
AH=12H

Exit AL=status
AH=extra status

Description Checks the status of the modifier keys and returns the result as separate bits in AL and AH, as follows:

AL	Significance
D7=1	INS mode
D6=1	CAPS LOCK mode
D5=1	NUM LOCK mode
D4=1	SCROLL LOCK mode
D3=1	[Alt] pressed
D2-1	[Ctrl] pressed
D1=1	left [Shift] pressed
D0=1	right [Shift] pressed

AH	Significance
D7=1	[Sys Req] pressed (always 0)
D6=1	CAPS LOCK mode
D5=1	NUM LOCK mode
D4=1	SCROLL LOCK mode
D3=1	right [Alt] pressed (always 0)
D2-1	right [Ctrl] pressed (always 0)
D1=1	left [Alt] pressed
D0=1	left [Ctrl] pressed

Set temporary shift (+)

Entry AH=20H
AL=shift status

Exit Returns nothing

Description Sets the shift status as indicated by AL:

D4=1	Scroll lock on
D5=1	Num lock on
D6=1	Caps lock on

The user can override this by pressing the appropriate keys.

Control shift keys (+)

Entry

AH=21H
AL=keys to control
BL=shift state required

Exit

Returns nothing

Description

Disables shift keys and sets them to a specified state. The user cannot state of disabled keys.

In both AL and BL, individual bits correspond to different keys, as follows:

Bit	Key
7	[Ins]
6	[Caps Lock]
5	[Num Lock]
4	[Scroll Lock]
3	[Alt]
2	[Ctrl]
1	Left [Shift]
0	Right [Shift]

Bits set in AL specify keys to disable. Bits set in BL specify the state to set those disabled keys in.

Control [Ctrl] [Alt] [DEL] (+)

Entry

AH=21H
AL=control (0=enable, non-0=disable)
BH=7
BL=8
CH=9
CL=10H
DH=11H
DL=12H

Exit

AL=0 (success)

Description

Disables or re-enables [Ctrl] [Ait] [DEL], the key combination that usually reboots an MS-DOS computer.

Use this function to prevent a user breaking out of an application unless they know about the emergency breakout sequence. (But also See function 23H).

The other registers must be set as shown for the function to operate. This should prevent accidental operation.

Control emergency breakout (+)

Entry

AH=23H
AL=control (0-enable, non-0=disable)
BH=7
BL=8
CH=9
CL=10H
DH=11H
DL=12H

Exit

AL-0 (success).

Description

If the user presses the [X] and [P] keys while turning on, **Hunter 16** will usually reboot even if it is locked into a program. This is the emergency breakout mechanism.

Use this function to disable emergency breakout, to prevent a user breaking out of an application. You should also use function 22H to prevent use of [Ctrl] [Alt] [DEL].

The other registers must be set as shown for the function to operate. This should prevent accidental operation.

WARNING: If an application locks up with both emergency breakout and [Ctrl] [Alt] [DEL] disabled, you will need to contact your supplier for assistance. (Or remove the main batteries and wait about several weeks until all power is exhausted from the internal backup battery, which will lose all data from the RAM disk!)

Redefine key codes (+)

Entry

AH=24H
AL=matrix code of key
BL=new key code

Exit

AL=0 (success) or AL<>0 (failed due to out of range parameters).

Description

Changes the key code generated by typing the key specified by AL. Each key code is then translated by the INT 9 interrupt handler, into a scan code (which is inserted in the keyboard buffer) or into an immediate action (such as toggle caps lock or turn off).

The matrix codes used to identify the keys are summarised in the following table.

Key codes and key scan codes are defined in Appendix E.

The standard key matrix is restored on a cold start. You can also restore the default keyboard by using function 25H.

Key matrix codes

Code	Key	Code	Key	Code	Key
0	[ESC]	30	[space]	60	[.]
1	[1]	31		61	
2	[Q]	32		62	Right [Shift]
3	[TAB right]	33	[0]	63	
4	[Num][Lock]	34	[-]	64	
5	[\]	35	[‘]	65	
6		36	keypad [7]	66	[8]
7		37	[RETURN]	67	[7]
8	Left [Shift]	38	keypad [7]	68	[U]
9	[Ctrl]	39		69	[I]
10	[PAW]	40	keypad [1]	70	[J]
11	[2]	41	keypad [0]	71	[K]
12	[W]	42		72	[M]
13	[A]	43		73	[N]
14	[S]	44	[=]	74	[/]
15	[Z]	45	[TAB left]	75	
16		46	keypad [8]	76	
17	[Alt]	47	keypad [9]	77	[6]
18		48	keypad [5]	78	[5]
19		49	keypad [6]	79	[T]
20		50	keypad [2]	80	[Y]
21		51	keypad [3]	81	[G]
22	[4]	52	keypad [.]	82	[H]
23	[3]	53		83	[B]
24	[E]	54		84	[V]
25	[R]	55	[9]	85	[#]
26	[D]	56	[O]	86	
27	[F]	57	[P]	87	
28	[X]	58	[;]	88	[PWR]
29	[C]	59	[L]	89	[Shift][PWR]

Reset keyboard (+)

Entry AH=25H

Exit AL=0 (success) as cannot fail.

Description Restores the normal IBM-compatible keyboard layout. This will mainly be used after the keyboard has been redefined by using function 24H.

The standard keyboard layout is also restored by a cold start.

Control keyclick (+)

Entry AH=26H
AL=control (0=disable, 1=enable)

Exit AL=0 (success) as cannot fail.

Description Note that keyclicks will also be disabled if sound output is disabled by using INT 15H function 70H, or the MUTE key ([PAW] [Shift] [Ctrl]).

The default is for keys to click, which is restored by a cold start.

Control screen dump area (+)

Entry AH=27H
AL=control (0=whole (virtual) screen, 1=LCD window)

Exit AL=0 (success) as cannot fail.

Description Controls the area of the screen that will be output when the screen is dumped by using INT 5 or pressing [Prtsc].

The default is to dump the whole window, which is restored on a cold start.

Get key repeat (+)

Entry AH=29H

Exit BL=typematic rate (0..1FH)
BH=delay (0..3)

Description Gets the delay before keys repeat, and the rate at which they repeat.

The rate (in characters per seconds) is given by BL:

BL	Rate	BL	Rate	BL	Rate	BL	Rate
0	30.0	8	15.0	10	7.5	18	3.7
1	26.7	9	13.3	11	6.7	19	3.3
2	24.0	A	12.0	12	6.0	1A	3.0
3	21.8	B	10.9	13	5.5	1B	2.7
4	20.0	C	10.0	14	5.0	1C	2.5
5	18.5	D	9.2	15	4.6	1D	2.3
6	17.1	E	8.6	16	4.3	1E	2.1
7	16.0	F	8.0	17	4.0	1F	2.0

The delay before a key repeats is set by BH:

BH	Delay
0	250mSec
1	500mSec
2	750mSec
3	1000mSec
4..FFH	Reserved

Control key repeat (+)

Entry AH=2AH
AL=control (0=disable, 1=enable)

Exit Returns nothing

Description Controls the keyboard autorepeat facility.

The default is for keys to repeat. This default is restored by a cold start.

Redefine key scan codes (+)

Entry AH=2BH
AL=0 (unshifted table), 1 (shifted table) or 2

(numlock table)
BH=standard scan code of key (0..128)
BL=new scan code to assign

Exit Returns nothing

Description Changes the scan code generated for each standard key (see function 24H). You can rede fine as many keys as you want and define the same code for several keys.

Note that the scan codes are used, NOT the ASCII codes, so you can redefine any key to any standard function.

For example, you could define the left [Shift] key as [Prtsc].

Key scan codes are defined in Appendix E.

The standard key codes are restored on a cold start. You can also restore the default keyboard by using function 25H, or redefine the keys at a lower level by using function 24H.

Redefine PAW key codes (+)

Entry AH=2CH
AL=matrix code of key
BL=new key code

Exit AL=0 (success) or AL<>0 (failed due to out of range parameters).

Description Changes the key code generated by typing the key specified by AL when the [PAW] key is held down. Each key code is then translated by the INT 9 interrupt handler, into a scan code (which is inserted in the keyboard buffer) or into an immediate action (such as toggle caps lock or turn off).

The matrix codes used to identify the keys are described under function 24H. Key codes and key scan codes are defined in Appendix E.

The standard key matrix is restored on a cold start. You can also restore the default keyboard by using function 25H.

Control Break keys (+)

Entry

AH=2DH
AL=mask
BX=0708H
CX=0910H
DX=1112H

Exit

AL=0 (success) or AL=FFH (failed)

Description

You can use this function to disable the Break keys and so prevent users breaking out of an application. AL is bit-significant, as follows:

AL	Effect
D0=0	Ignore [Ctrl-C]
D0=1	Normal [Ctrl-C] action
D1=0	Ignore [Ctrl-Break]
D1=1	Normal [Ctrl-Break] action

Normal operation of the Break keys is restored by a cold start.